

Astrophotographic Post-Capture Alignment

An Earth-Based Computer Vision Approach to Aligning Lunar Frames

Chase Kusterer^{1, †}

1. Northwestern University

[†] Email: chase.kusterer@faculty.hult.edu

[†] LinkedIn: <https://www.linkedin.com/in/kusterer/>

Problem Statement and Objectives

Amateur astrophotography often pairs an entry-level telescope with a professional camera body, using the telescope as the lens. Such configurations suffer from image degradation caused by equipment jitter and atmospheric turbulence. While hardware such as an equatorial tracking mount can reduce physical instability, this remains costly and cannot remove the blur introduced by shifting air. Atmospheric blur can instead be reduced by lucky imaging, which records many short-exposure frames and stacks the sharpest of them into one composite. However, lucky imaging depends on the target being closely aligned across all frames, which is difficult to achieve manually or with automated techniques such as gradient-based alignment. The objective of this project is to develop an alignment solution that also alleviates the effects of equipment jitter using modern computer vision techniques. More specifically, its architecture utilizes segment anything models (SAMs) and Laplacian-of-Gaussian blob detection (LoG) to align images of the Moon based on detected craters in eight highly visible maria.

Instrumentation and Data Collection

Images were captured using a Sony Alpha 7 IV mirrorless camera attached to a Celestron StarSense Explorer DX 130AZ Newtonian telescope. Configuration also included a 2x Barlow lens, T-adapter, and 1.25 inch focuser, leading to a magnification level of 39x, which is equivalent to observing the Moon from approximately 9,900 kilometers away. Furthermore, this configuration led to a pixel width of 1.50 km for images and 1.80 km for 4K videos, making craters and many other of the Moon's surface details visible. Focus needed to be controlled manually via the telescope's focus knobs as electronic camera lenses were incompatible with this configuration. The camera was also operated in APS-C crop mode with image stabilization disabled. Lunar frames were collected at varying ISO levels with shutter speeds between 1/125 and 1/500 of a second to control for atmospheric turbulence in each individual frame. Collection took place at the summit of Mount Diablo in California as well as locations in the surrounding area from August 4 through August 22, 2026.

Dataset Description and Preprocessing Approach

In order to achieve crater alignment between lunar images, a crater reference mask was engineered by developing a coordinate reference system (CRS) from a mosaic of 1,300 nearside wide angle images captured from the Lunar Reconnaissance Orbiter Camera (LROC) (NASA 2010) mapped to the Robbins Lunar Crater Database (Robbins 2019). This also resulted in a ground truth lunar reference image with a surface radius of 7800 pixels at a scale of approximately 0.22 kilometers per pixel. Crater scope was reduced to eight bounding boxes covering highly visible maria, which are vast planes of solidified basaltic lava once thought to be seas. A total of 295 craters were mapped and masked, covering 3.80 percent of the reference frame (see Appendix A1). More information on Mare bounding boxes can be found in Appendix B.

The test frame utilized during development was captured during a waning gibbous phase of 59 percent illumination. It was preprocessed to a 16-bit TIF format where its three color channels were averaged into one. Additionally, as the test frame was in landscape format, it was rotated 90 degrees clockwise for better orientation with the crater reference mask (see Appendix A2).

Mathematical Foundations and Model Architectures

After developing a crater reference mask, binary lunar disk segmentation was conducted on the test frame and the reference image to separate the Moon from its respective dark sky. The base Segment Anything Model (SAM) performed exceptionally well due to the intense contrast between the object of interest and its background. This model, ViT-B, contains approximately 91 million parameters, making it the smallest and fastest of Meta's SAM models (Kirillov 2023). During development, ViT-B would occasionally over segment the reference image, correctly masking the circular shape of the Moon but leaving a few pinholes on the disk body. To mitigate this risk, a post-segmentation step was implemented to clean each mask by identifying its largest piece and filling any interior holes.

A random sample consensus (RANSAC) circle was then fit around each segment based on its limb (i.e., the circular edge of the near side of the Moon). This was done to estimate the unlit portion of the Moon in the test frame so

that it could be scaled to the reference image. The RANSAC circle for the reference image fit extremely well at 7744 pixels, or 99.28 percent of its segment's surface radius. Then, by utilizing the circumferences of the RANSAC circles, the test frame was resized and translated to align with the reference image, resulting in the test frame scaling by a factor of 1.425 (see Appendix A3). After reviewing the overlap between the images, initial scaling and translation were both successful and are ready for fine tuning. A small translation search was conducted to maximize the overlap between the two segments. An overlap of 59% would be ideal given that this was the Moon's illumination level when the test frame was captured.

Training Methodology and Hyperparameter Choices

After scale and translation had been addressed, rotation was the remaining a factor to align the images further. Craters in the test frame were detected using a centroid-based non-buffered Laplacian-of-Gaussian blob detector (LoG) and evaluated against the crater reference mask. LoG is an advanced computer vision algorithm originally designed to predict circular structures, which made it a strong candidate for this task. Four hyperparameters were engineered and tuned to control the number of craters being utilized for alignment, crater diameter minimums and maximums, and the crater magnification level. Using a Bayesian tuning approach with Optuna over 500 trials, an optimal set of hyperparameters was discovered.

Evaluation Metrics, Results, and Benchmarks

The DICE coefficient and intersection over union (IoU) were utilized in evaluating alignment performance. Binary segmentation disk alignment resulted in an intersection over union (IoU) value of 0.58, implying an illumination level of 58% (98.3% of the reference illumination benchmark stated earlier, see Appendix A4). For centroid-based crater alignment, a DICE coefficient of 0.037 was established as a benchmark using the nearest 295 craters with no size limitations or magnification. As expected, this value was surpassed with hyperparameter tuning, which yielded a DICE coefficient of 0.898 with the optimal configuration consisting of the two closest craters with diameters between 30 km and 203 km and a 2x magnification level. Additional rotation tuning to optimize the alignment between the test frame and the crater mask led to a final DICE coefficient 0.909, approximately twenty five times more than the established benchmark (see Appendix A5).

Comparison with Alternative Approaches

This project originally applied a transfer learning approach using the DeepMoon U-Net architecture (Silburt 2018), but it proved to be a very poor fit for celestial alignment. Therefore, AutoStakkert!, the current leading open source astrophotography post-capture software (Kraaikamp n.d.) was utilized for comparison. AutoStakkert!'s primary purpose is stacking video frames for lucky imaging, which includes an alignment component. A limitation of its alignment approach is that the software automatically creates a reference frame based on the frames provided instead of allowing a user to import one. Thus, it is infeasible to evaluate results based on ground truth. Furthermore, AutoStakkert! does not return a DICE coefficient or Intersection over Union (IoU) value to assess alignment quality.

Challenges Encountered and Solutions Implemented

Two major challenges were encountered in this project. First, the coordinate reference system (CRS) that was developed and utilized in this analysis was modeled as a flat disk with the Moon's sub-Earth point (i.e., the central point of the near side of the Moon) as the origin. Accuracy was high for coordinates near the sub-Earth point, but it degraded close to the limb. To alleviate this, CRS scope was reduced to eight central and highly visible maria.

Second, DeepMoon was trained on a global digital elevation map (Silburt 2018) which led to negative transfer on the pixel-based image captures collected for this project. This challenge was alleviated by pivoting to a segmentation approach with the lightweight SAM ViT-B model. This change proved successful for disk segmentation, a key component of the project's pipeline.

References

1. Kirillov, A., Mintun, E., Ravi, N., Mao, H., Rolland, C., Gustafson, L., ... & Girshick, R. (2023). Segment Anything [Computer software]. GitHub. <https://github.com/facebookresearch/segment-anything>
2. Kraaikamp, E. (n.d.). *Features and guides*. AutoStakkert! Retrieved August 30, 2026, from <https://www.autostakkert.com/wp/guides/>
3. NASA, GSFC, Arizona State University. (2010). *Nearside WAC mosaic* [Image]. Lunar Reconnaissance Orbiter Camera. <https://lroc.im-ldi.com/visit/exhibits/1/gallery/17>
4. Robbins, S. J. (2019). A new global database of lunar impact craters greater than 1 to 2 km. *Journal of Geophysical Research: Planets*, 124(4), 871-892.
5. Silburt, A., & Zhu, C. (2018). *DeepMoon* [Computer software]. GitHub. <https://github.com/silburt/DeepMoon>

Appendix A. Project Visualizations

Appendix A1

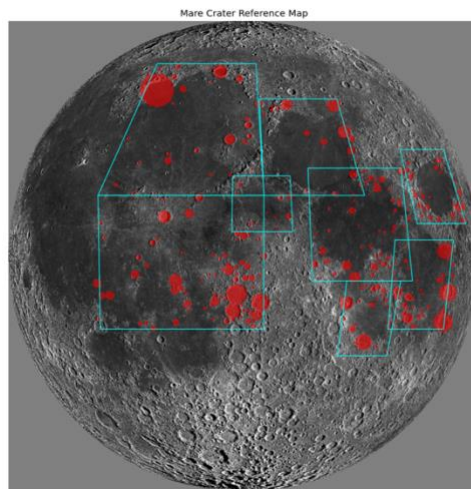


Figure A1: Mare crater reference map developed from a LROC mosaic and the Robbins Lunar Crater Database.

Appendix A2



Figure A2: Preprocessed test frame, captured August 5, 2026 at 12:48am (PST)

Appendix A3



Figure A3: Segmented test frame (top left), segmented reference image (top right), test frame RANSAC circle (bottom left), reference image RANSAC circle (bottom right)

Appendix A4

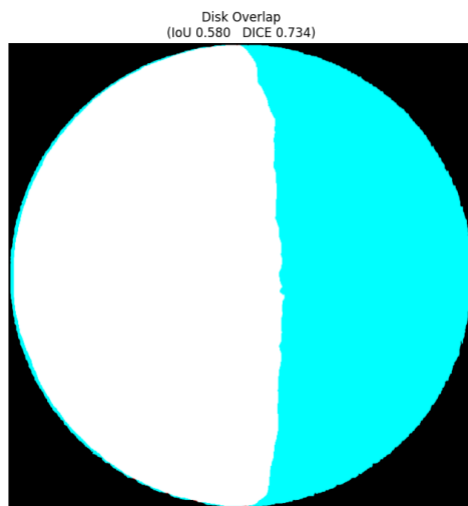


Figure A4: Disk overlap between the test frame and reference image.

Appendix A5

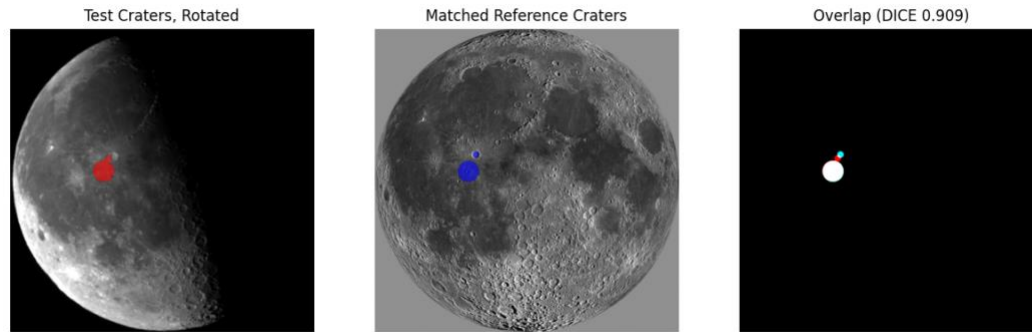


Figure A5: Rotation alignment: test craters (left), matched reference craters (center), and segmentation overlap (right).

Appendix B. Mare Bounding Boxes

Crater scope bounding boxes covering highly visible maria.

Mare	Latitude	Longitude (East)
Imbrium	15 to 55	-40 to 5
Serenitatis	15 to 42	5 to 32
Tranquillitatis	-6 to 22	17 to 46
Vaporum	6 to 20	-3 to 12
Nectaris	-25 to -6	26 to 42
Crisium	8 to 27	48 to 72
Fecunditatis	-18 to 4	40 to 63
Imbrium to Humorum	-18 to 15	-40 to 5

Table B1: Latitude and longitude (east) for the eight mare bounding boxes.